



GIET UNIVERSITY GUNUPUR-765022

SCHOOL OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CSE-AIML

Assignment

Assignment No: 3
Subject Code: 25BBSBS23004
Semester : 3rd
Date of Assignment: 29/09/2026
Submission Date : on or before
5/10/2026

Subject Name: Applied Statistics
Academic Year: 2026-2027
Prog. & Course: B. Tech CSE-AIML
Section(s): A, C and E
Faculty Name: Dr. Boina Anil Kumar

1. In a frequency distribution the coefficient of skewness based on quartiles is 0.6. If the sum of the upper and lower quartiles is 100 and median is 38, find the value of upper and lower quartiles.
2. For a distribution the distance of the median from the first quartile is five times the distance of the third quartile from the median. Calculate a measure of skewness for this distribution. Give a rough sketch of the distribution.
3. Calculate the first four central moments about the mean for the following frequency distribution. Also, comment on the nature of the distribution.

x	0	1	2	3	4	5	6	7
f	2	5	10	16	20	15	8	4

4. Plot a Scatter Plot for the following heights and weights of 10 students.

Height	62	72	68	58	65	70	66	63	60	72
Weight	50	65	63	50	54	60	61	55	54	65

5. The marks obtained by six students in Mathematics and Statistics are given below. Calculate Karl Pearson's coefficient of correlation.

Mathematics (X)	40	45	50	55	60	65
Statistics (Y)	42	48	46	55	58	62

6. The marks obtained by 8 students in Mathematics and Statistics are given below.

Student	A	B	C	D	E	F	G	H
Mathematics	85	75	90	75	60	68	55	55
Statistics	78	82	90	70	65	75	60	60

Calculate Spearman's Rank Correlation Coefficient.

7. Fit the regression line of X on Y for the following data:

X	10	20	30	40	50
Y	12	18	25	30	35

Estimate X when $Y = 28$.

8. Fit the regression line of Y on X for the following data:

X	2	4	6	8	10
Y	3	5	7	8	10

Also estimate Y when $X = 12$.

9. If t is an unbiased estimator of the population mean μ , where

$$t = \frac{2X_1 + X_2 + \lambda X_3}{5},$$

find the value of λ .

10. Let T_1 and T_2 be consistent estimators of μ_1 and μ_2 respectively. Prove that

$$aT_1 + bT_2$$

is a consistent estimator of

$$a\mu_1 + b\mu_2.$$

11. If X_1, X_2, X_3 constitute a random sample of size 3 from a normal population with population mean μ and variance σ^2 , find the efficiency of

$$\frac{X_1 + 2X_2 + X_3}{4}$$

relative to

$$\frac{X_1 + X_2 + X_3}{3}.$$

12. Find the maximum likelihood estimators (MLEs) of μ and σ^2 for Normal distribution.
13. A sample of 36 observations from a normal population has a sample mean of 105. The population standard deviation is known to be 12. Find the 99% confidence interval for the population mean μ .
14. The following observations are obtained from a normal population:

$$12, \quad 15, \quad 14, \quad 10, \quad 13, \quad 16, \quad 11, \quad 14.$$

Find the 95% confidence interval for the population mean μ .

15. A population has a known proportion $P = 0.60$ of defective items. A random sample of 400 items is selected from the population, and the sample proportion is found to be

$$p = 0.56.$$

Find the 95% confidence interval for the population proportion P .

Instructions

- Use A4 size papers to write the answers.
- Write legibly with proper alignment and margins.
- Answer all the question in a proper order.
- Submit your assignment on or before the due date.
- The first page of your assignment should include the following information:

Assignment No-

Name :
Roll No :
Semester :
Course :
Section :
Branch :
Subject :
Academic Year :
Date of Submission :

Dr. Boina Anil Kumar